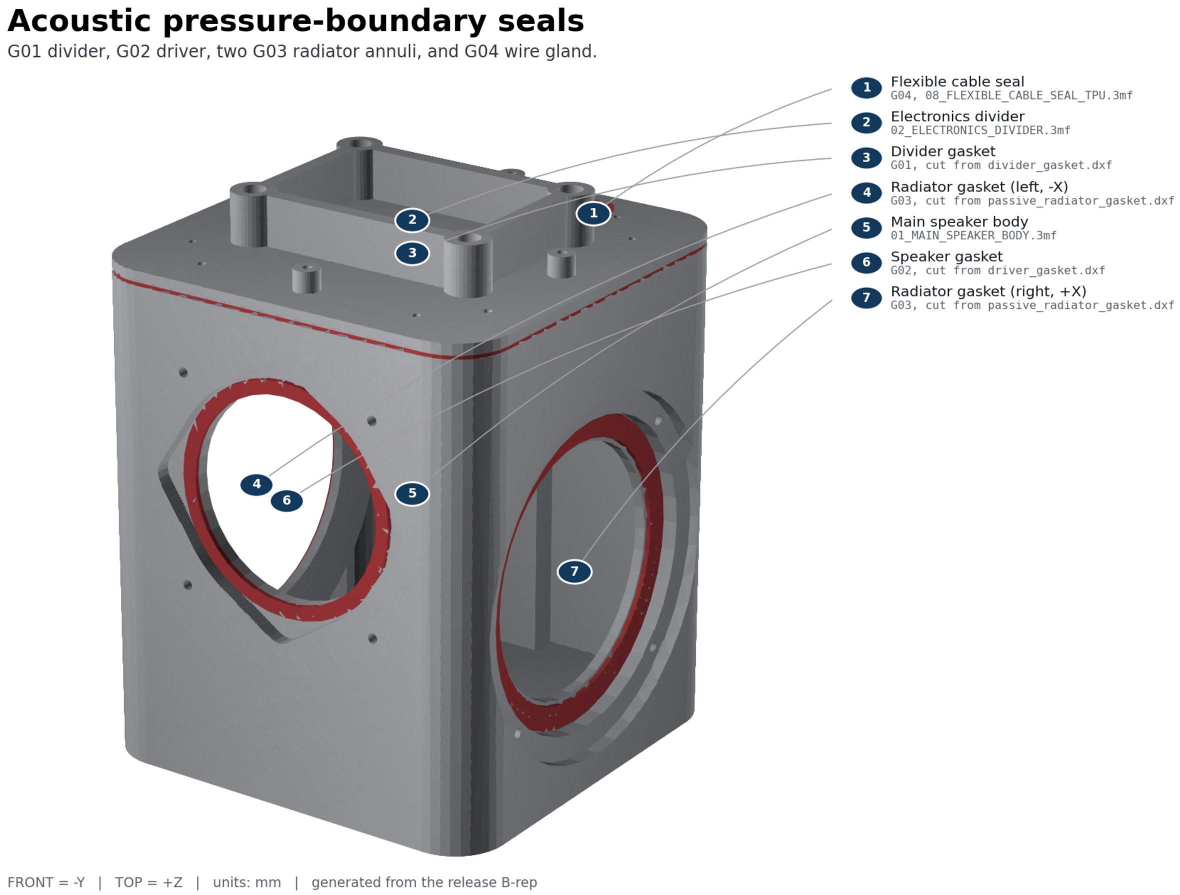
Testing and Commissioning Guide

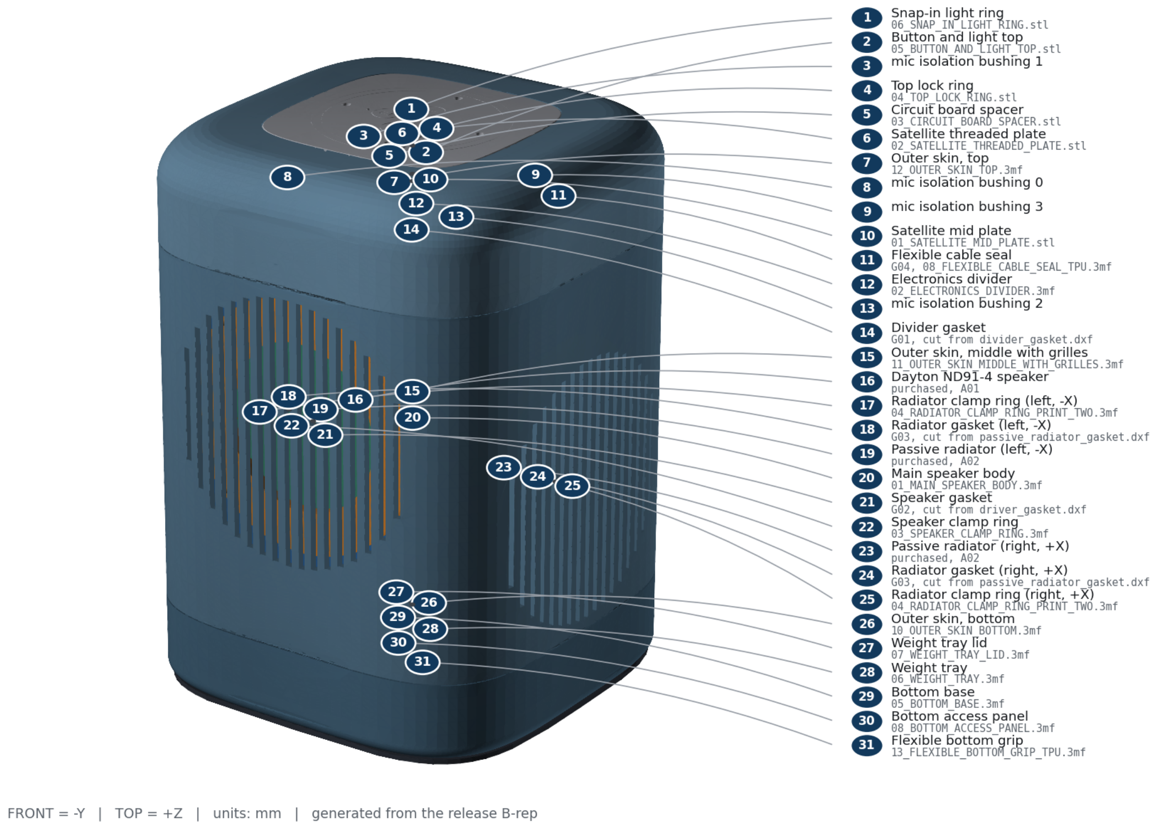
Every result in this guide requires a physical unit and is `REQUIRES_PHYSICAL_VALIDATION`.



Seal locations

STEP 9 **Fit the anti-slip ring and complete inspection**

Final inspection: all seams even, all openings clear, no loose hardware.



Final assembled inspection

Before power

1. Verify red driver lead to + and black to -.
2. Confirm both radiators carry equal added mass: model target 7.99 g each.
3. Confirm every screw ID and quantity against `FASTENERS.csv`.
4. Confirm G01-G04 are continuous and no wire touches a moving component.
5. Perform the 100-250 Pa gross-leak screen during assembly. Never use shop air.

Controlled first power

Use a current-limited supported USB-C supply and the official Batch 1 firmware. Start muted, then at minimum volume.

• LEDs: all segments visible and even.

• Buttons: each click registers once and returns freely.

• USB-C: plug inserts/removes without shell contact.

• Wi-Fi: connect and record RSSI beside a bare Batch 1 control.

• Microphones: verify all four channels, then run 50 wake-word trials at 1, 3, and 5 m on-axis and 45 degrees.

• Audio: play a polarity pulse, then a 100-500 Hz sweep at low level. Stop for rub, buzz, air noise, or asymmetric radiator motion.

Acoustic commissioning

Measure impedance magnitude and phase from 20 Hz to 20 kHz at low level. The two low-frequency peaks should bracket tuning; the minimum between them is the real Fb. Model targets are 60.0 Hz Fb, 4.39 ohm minimum impedance, and 59.0 Hz f3. They are `ENGINEERING_ESTIMATE`, not pass/fail measurements.

Start DSP with the modelled 51.0 Hz fourth-order high-pass and no positive bass boost. Final EQ, limiter, and tuning mass require measured response and excursion.

Thermal and reliability

Instrument Core SoC, amplifier area, and electronics-bay air. Test 60 minutes idle and 60 minutes pink noise at 25 C, then repeat at 35 C. Pass only if every supplier limit retains 15 C margin, there is no throttling, and no printed part softens. Perform a gentle rattle check and repeat the leak/impedance check after five service cycles.

Record results; do not change the repository status to physically validated without the measurements.

Source commit aceaa4b95133. No physical validation has been performed.